

18-427/727: Board-Level RF Systems for IoT

LAB01: Transmission Lines & VNAs

Writeup Due: Friday, September 19th, 2025

How to submit labs:

Create a word document or PDF that answers the questions presented below. Include any screenshots or figures as requested. **Please provide a descriptive caption for each figure!**

Question:	1	2	3	4	Total
Points:	15	35	25	25	100
Score:					

Lab Outline

This lab aims to introduce the Vector Network Analyzer, an essential tool for studying RF systems. With this device, we will also perform characterization of a transmission line stub, as well as a patch antenna and a t-junction power splitter. You will be given a set of boards for your team to use during the lab.

It may be helpful to view this video on [wave theory](#) from Bell Labs before or shortly after completing this lab session. This will complement your experience in the lab by helping you visualize the phenomena of waves traveling down transmission lines, as well as how waves are reflected and impedance matching. These concepts are fundamental not only to this course but to countless fields in electrical engineering, physics, and more!

1. Calibrating the VNA
2. Transmission line stub
3. Patch antenna
4. T-Junction

Equipment Required

- LibreVNA - network analyzer tool (provided by instructor)
- Set of boards - microstrip trace, patch antenna, and power splitter (provided by instructors)

1. Calibrating the VNA

Before you get started, you'll need to download the LibreVNA software on your laptop. The instructions below advise you on installing the LibreVNA software on your own machine:

In order to use the VNA, we must first install the LibreVNA software and then calibrate the device. [This link](#) takes you to the Github repository for the LibreVNA. There is a lot of documentation on the repository, but we're mostly interested in the [Quick Start](#) section, which has a link to the [GUI installers](#) and the [user manual](#). For the GUI installer, the most recent update is v1.4.0 as of the writing of this lab manual; there should be options for Windows, Mac, and Ubuntu. Download, unzip, and run the appropriate installer for your OS. If you have any installation issues, please reach out to the TAs for assistance.

With the GUI running, we can get started with the user manual. It's worth taking a look at the entire manual to familiarize yourself with what this VNA can do, but for now, you can skip to Section 3.5 in the user manual. This will show us how to calibrate our VNA by using the SOLT calibration type (short, open, load). **Set your calibration sweep from 300MHz to 3GHz.**

Please set up the GUI to look like the following:

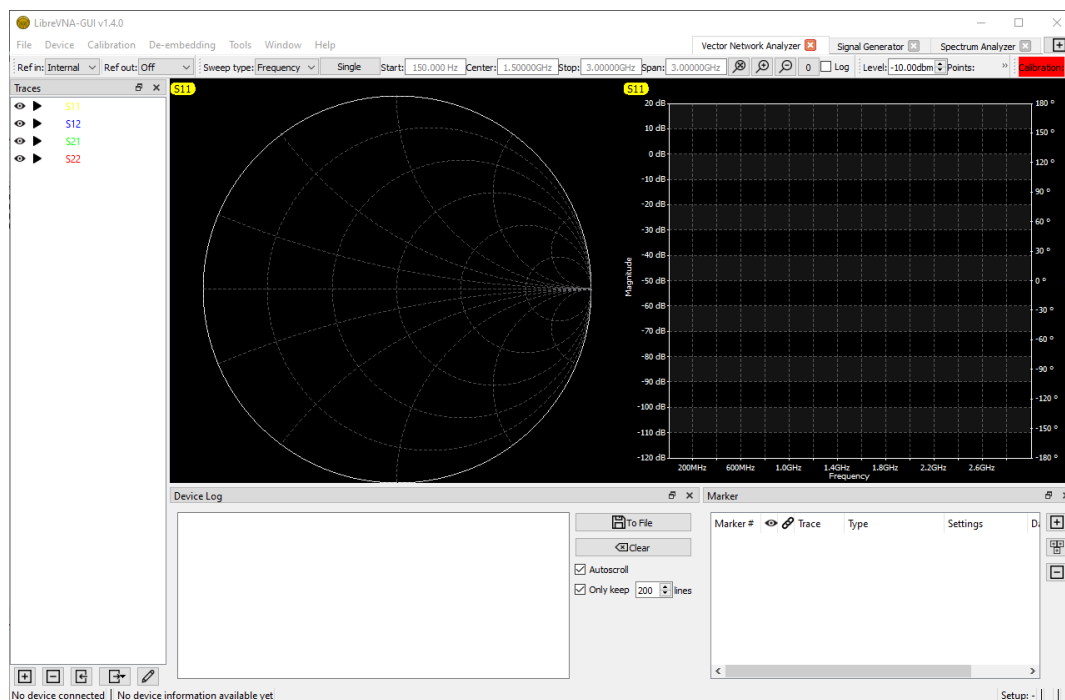


Figure 1: LibreVNA GUI

S11 should be plotted twice: on a Smith Chart on the left, and on an XY Plot on the right. Right-click unwanted graphs and select Close, then click Close Tile to remove them. Make sure both plots are for S11 by right-clicking, then selecting S11.

15 pts

1.1 Include a screenshot of each of these measurements after calibration:

- S11, Port 1 terminated with open
- S11, Port 1 terminated with short
- S11, Port 1 terminated with load

2. Transmission Line Stub

You have been provided with a segment of transmission line that has **two ports**. As you may recall from lecture, transmission line theory is critical in designing RF circuits, as the properties and dimensions of the material will affect the propagation of the electromagnetic waves. As RF circuits operate in the hundreds of MHz and GHz, we need to carefully design our boards to connect transceivers to antennas and other devices.

For this lab, we have designed a set of boards for you to investigate using the VNA. The first board is the transmission line, which we have implemented as a **microstrip** (as opposed to a coplanar waveguide or other such structure). This means that there is a conductor - the copper strip - on top of a dielectric substrate (FR4), which is above a copper ground plane. We have used FR4 as the substrate, which is commonly used in printed circuit boards. Rogers, another substrate material, has a dielectric material that is more suitable for high frequencies, though it is more expensive.



Figure 2: Microstrip transmission line segment.

Now, you will notice that the transmission line has two ports, which are SMA connectors. Connect one end to your VNA, making sure that you have calibrated the system as described in Part 1 of this lab. Using the other port, connect the open, short, and 50 Ohm loads. With each load connected, you will do a sweep in the VNA software tool. This will show you the behavior of the system on the Smith chart; sweep from 500MHz to 1GHz to get a good idea of the impedance.

15 pts

- 2.1** Include screenshots of your open-, short-, and load-terminated sweeps for S11. **Sweep from 500MHz to 1GHz.**

10 pts

- 2.2** What is the frequency at which this stub acts like a quarter-wave transformer? Recall that for a quarter-wave transformer,

$$Z_{in} = \frac{Z_0^2}{Z_L}$$

Justify your answer.

10 pts

- 2.3** For transmission lines on a dielectric substrate (see Eq 4.12 in Steer):

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_e}}$$

λ_0 is the free-space wavelength, λ_g is the effective wavelength in the transmission line, and ϵ_e is the effective dielectric constant of the transmission line. **The transmission lines we have provided you are about 5.35 cm in length.**

What is your estimate of the effective dielectric constant of this microstrip based on your previous answer? Justify your answer.

3. Patch Antenna

Now that we have taken some basic measurements of a transmission line, we can begin to study one of the most important parts of an RF system: the antenna. Patch antennas are a very useful design for RF transceivers due to their compact form, fitting conveniently on a PCB. These types of antennas will be used frequently throughout this course.

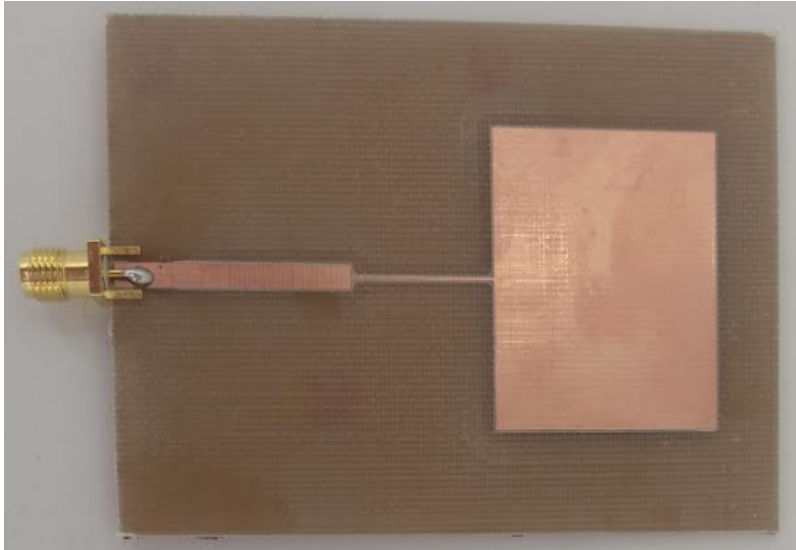


Figure 3: Patch antenna

In this lab, we will take a look at a patch antenna designed to resonate at a particular frequency. Antennas typically have a single port, because the energy entering the antenna is intended to radiate away! Connect your VNA to the patch antenna's port so that you may characterize the antenna.

10 pts

3.1 Include a screenshot of your plot of S11. **Sweep from 2 GHz to 3 GHz.**

5 pts

3.2 What is the center frequency for this patch antenna?

10 pts

3.3 What is the VSWR at this center frequency? Recall that

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

4. T-Junction

In the last part of this lab, you will take measurements of a T-junction power splitter, as shown below. **Measure S11 with the other two ports terminated with 50 Ohm loads.** Ideally, the power incident on Port 1 would be divided and transferred to the two loads, leaving no reflection. However, this is a naive splitter design that does not function the way we would hope.

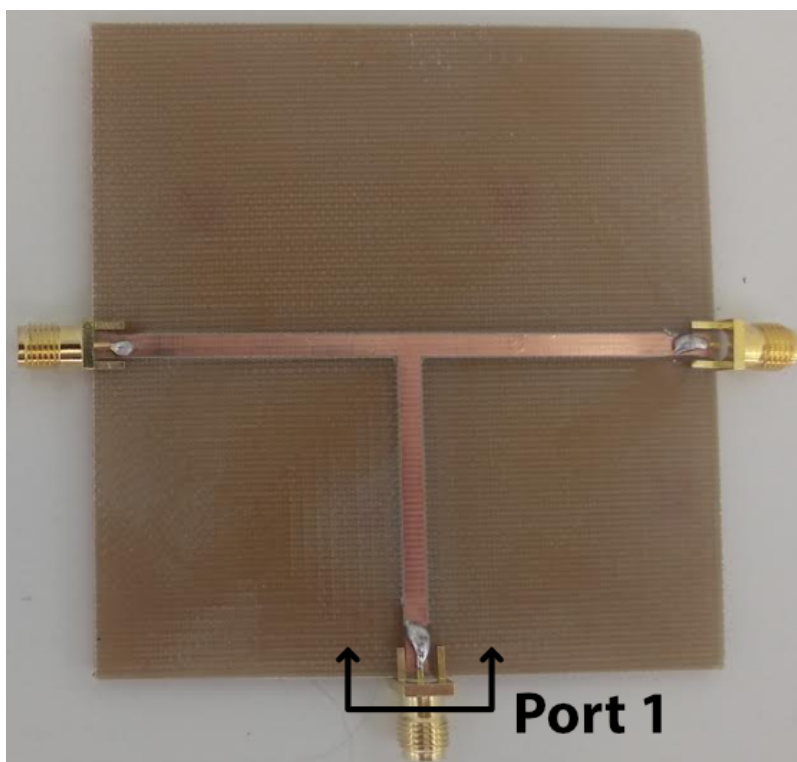


Figure 4: T-Junction

5 pts

4.1 Include a screenshot of your plot of S11. Sweep from 500 MHz to 1.5 GHz.

5 pts

4.2 What did you measure for $|\Gamma|$, the magnitude of the voltage reflection coefficient, in absolute units (i.e. not dB)?

10 pts

4.3 Why is there a reflection, even with the two output ports terminated with 50 Ohm loads?

5 pts

4.4 How should the transmission lines going to the output ports be modified to improve the match (i.e. reduce $|\Gamma|$) of this power splitter?

Reflections

Please complete our [post-lab survey](#) for 5 points of extra credit! Attach a photo of the form completion page to your lab submission.